

4in (10cm) tall, yellow flowers open flat. Z10–11 (min. 41°F/5°C)

P. mammulosa, syn. *Notocactus mammulosus*. Spherical, perennial cactus. H 4in (10cm), S 2½in (6cm). Green stem has 13–21 ribs and straight, stiff, yellow-brown to white spines, to ½in (1cm) long. Woolly crown produces abundant, golden flowers in summer. Z10–11 (min. 41°F/5°C)

P. microsperma, syn. *P. mutabilis*, *P. sanguiniflora*. Clump-forming, perennial cactus. H 3in (8cm), S 1ft (30cm). Has a much-ribbed, green stem densely covered with brown, radial spines and red, central spines, some of which are hooked. Bears blood-red, occasionally yellow, flowers in spring. Z11–12 (min. 50°F/10°C)

P. mueller-melchersii, syn. *Notocactus rutilans*, *P. rutilans*, illus. p.486.

P. mutabilis. See *P. microsperma*.

P. nivosa illus. p.487.

P. ottonis, syn. *Notocactus ottonis*. Variable, spherical, perennial cactus. H 1¼–6in (3–15cm), S 7in (18cm). Has a pale to dark green stem with 6–15 rounded ribs bearing stiff, golden radial spines and longer, soft, red central spines. In summer, crown bears flattish, glossy, golden flowers, 3in (8cm) across, with purple stigmas. Offsets freely from stolons. Prefers sun. Z10–11 (min. 41°F/5°C)

P. rutilans. See *P. mueller-melchersii*.

P. sanguiniflora. See *P. microsperma*.

P. scopa, syn. *Notocactus scopa* (Silver ball cactus). Spherical to columnar, perennial cactus. H 2–20in (5–50cm), S 4in (10cm). Stem, with 25–40 ribs, is densely covered with white radial spines and longer, red central spines, 3 or 4 per areole. Crown bears funnel-shaped, glossy, yellow flowers, 1½in (4cm) across, with purple stigmas, in summer. Prefers a sunny position. Z10–11 (min. 41°F/5°C)

PARONYCHIA

Whitlow-wort

CARYOPHYLLACEAE

Genus of annuals and evergreen perennials making loose mats of prostrate stems. Good carpeting plants for rock gardens and for walls. Fully to half hardy. Needs sun and well-drained soil. Propagate by division in spring.

P. capitata, syn. *P. nivea*. Vigorous, evergreen, mat-forming perennial. H to 2in (5cm), S 1ft (30cm). Silvery leaves are small and ovate. In summer, produces tiny flowers within papery bracts. Good groundcover. (borderline) Z5–8

P. kapela subsp. serpyllifolia illus. p.375.

P. nivea. See *P. capitata*.

PARROTIA

HAMAMELIDACEAE

Genus of deciduous trees, grown for their flowers and fall color. Fully hardy, but flower buds may be killed by hard frosts. Requires full sun and grows best in fertile, moist but well-drained soil. Lime-tolerant, but usually colors best in acidic soil. Propagate by softwood cuttings in summer or by seed in fall.

P. persica (Persian ironwood) illus. p.77.

‘Vanessa’ is an upright, columnar tree. H 25–30ft (8–10m), S 6–10ft (2–3m), Has

bronze to red-margined young leaves, turning crimson, purple, orange, and gold in fall. In late winter to early spring, bears dense, globular heads of flowers with conspicuous, red anthers. Good choice for small gardens. Z5–7

PARROTIOPSIS

HAMAMELIDACEAE

Genus of one species of deciduous shrub, grown for its ornamental, dense flower heads surrounded by conspicuous bracts. Fully hardy. Needs sun or partial shade. Grows in any fertile, well-drained soil except very shallow soil over chalk. Propagate by softwood cuttings in summer or by seed in fall.

P. jacquemontiana. Deciduous, upright shrub. H 20ft (6m), S 12ft (4m). Has witch-hazellike, dark green leaves that turn yellow in fall. From mid- to late spring and in summer, bears clusters of minute flowers, with tufts of yellow stamens, surrounded by white bracts. Z5–7

PARTHENOCISSUS

VITACEAE

Genus of deciduous, woody-stemmed, tendril climbers, grown for their leaves, which often turn beautiful colors in fall. Broad tips of tendrils have suckerlike pads that cling to supports. Has insignificant, greenish flowers in summer. Will quickly cover north- or east-facing walls or fences and may be grown up large trees. Fully to frost hardy. Grow in partial shade or shade and in well-drained soil. Propagate by softwood or greenwood cuttings in summer or by hardwood cuttings in early spring. ☉The berries may cause mild stomach upset if ingested.

P. henryana, syn. *Vitis henryana*. Deciduous, tendril climber with 4-angled, woody stems. H to 30ft (10m) or more. Leaves have 3–5 toothed, ovate leaflets, each to 5in (12cm) long, and are velvety, deep green, or bronze with white or pinkish veins. Small, dark blue berries are produced in fall. Leaf color is best with a shaded aspect. Z7–8

P. quinquefolia, syn. *Vitis quinquefolia* (Virginia creeper). Deciduous, woody-stemmed, tendril climber. H 50ft (15m) or more. Leaves have 5 ovate, toothed, dull green leaflets, paler beneath, that turn a beautiful crimson in fall. Bears blue-black berries in fall. Will cover a high wall or building, but can be invasive. Z3–9

P. striata. See *Cissus striata*.

P. thomsonii. See *Cayratia thomsonii*.

P. tricuspidata (Boston ivy, Japanese ivy) illus. p.209.

‘Lowii’ illus. p.209.

‘Veitchii’ (syn. *Ampelopsis veitchii*) illus. p.210.

PASSIFLORA

Passion flower

PASSIFLORACEAE

Genus of mostly evergreen, woody-stemmed, tendril climbers, grown for their unique flowers, each one with a central corona of filaments. Also includes a few annuals, perennials, shrubs, and trees. Produces egg-shaped to rounded, fleshy, edible fruits that mature to orange or

yellow in fall. Fully hardy to frost tender. Grow in full sun or partial shade and fertile, well-drained soil. Water freely in full growth, less at other times. Stems need support. Thin out crowded growth in spring. Propagate by seed in spring or by semiripe cuttings in summer.

P. alata. Robust climber. H to 20ft (6m) or more. Has square stems and broadly ovate, sometimes finely toothed, rich to light green leaves. From spring to late summer, pale crimson buds open to nodding, fragrant, bowl-shaped, bright carmine flowers; crowns have purple, red, and white zones. Yellow fruits are egg- to pear-shaped. Z10–12 (min. 41–45°F/5–7°C)

P. x allardii, syn. *P. ‘Allardii’*. Strong-growing, evergreen, woody-stemmed, tendril climber. H 22–30ft (7–10m). Has 3-lobed leaves. In summer–fall, bears white flowers, 3–4in (7–10cm) across, tinted pink, with purple-banded crowns. Z7–11

P. ‘Allardii’. See *P. x allardii*.

P. ‘Amethyst’, syn. *P. amethystina* of gardens, *P. ‘Lavender Lady’*, *P. violacea* of gardens. Vigorous climber. H 12ft (4m) or more. Has smooth, slender stems and deeply 3-lobed, rich green leaves. In late summer and fall, bears bowl-shaped, purple to purple-blue flowers with green anthers and darker crown filaments, followed by ovate, orange fruits. Z9–11

P. amethystina of gardens. See *P. ‘Amethyst’*.

P. caerulea (Common passion flower) illus. p.204.

‘Constance Elliot’ is a fast-growing, evergreen or semievergreen, woody-stemmed, tendril climber. H 30ft (10m) or more. Has rich green, deeply divided leaves. From summer to fall, produces bowl-shaped, fragrant, white flowers with pale blue or white filaments. Z7–9.

‘White Lightning’ has ivory-white flowers with violet-blue filaments.

P. x caeruleoracemosa. See *P. x violacea*.

P. x caponii ‘John Innes’. Strong-growing, evergreen, woody-stemmed, tendril climber. H 25ft (8m). Has 3-lobed leaves. Bears bowl-shaped, nodding, white flowers, flushed claret-purple, with purple-banded, white crowns, from summer to fall. Z12 (min. 59°F/15°C)

P. coccinea, syn. *P. fulgens*, *P. velutina* (Red grenadilla), illus. p.462.

P. ‘Damsel’s Delight’, syn. *P. ‘Silly Cow’*. Vigorous, evergreen or semievergreen, tendril climber. H 10–15ft (3–5m). From summer to fall, bears white to pale lilac flowers, 5in (12cm) across, with bright violet-blue filaments. Z8–11

P. x exoniensis. Fast-growing, evergreen, woody-stemmed, tendril climber. H 25ft (8m) or more. Leaves have 3 deep lobes and are softly downy. Rose-pink flowers, 3in (8cm) across, that have purple crowns with white teeth, are borne in summer to fall. Z10–11 (min. 41°F/5°C)

P. fulgens. See *P. coccinea*.

P. ‘Incense’. Vigorous climber. H to 10ft (3m). Has deeply lobed, rich green leaves. From late spring to early fall, bears lacy, fragrant, bowl-shaped, violet-mauve flowers, with crowns of wavy filaments, white near the bases. Egg-shaped, edible but acidic, yellow fruits are borne in summer and fall. Z10–11

P. ‘Lavender Lady’. See *P. ‘Amethyst’*.

P. manicata. Fast-growing, evergreen, woody-stemmed, tendril climber. H 10–15ft

(3–5m). Has slender, angular stems and 3-lobed leaves. Red flowers, with deep purple and white crowns, are borne from spring to fall. Z10–11 (min. 41°F/5°C)

P. quadrangularis (Giant grenadilla) illus. p.462.

P. racemosa (Red passion flower). Fast-growing, evergreen, tendril climber with slender, woody stems. H 15ft (5m). Has wavy, leathery leaves with 3 deep lobes. Bears clusters of pendent, crimson flowers, 3–4in (8–10cm) across, with white- and purple-banded crowns, in summer and fall. Z12 (min. 61°F/16°C)

P. sanguinea. See *P. vitifolia*.

P. ‘Silly Cow’. See *P. ‘Damsel’s Delight’*.

P. velutina. See *P. coccinea*.

P. x violacea, syn. *P. x caeruleo-racemosa*. Vigorous, evergreen, woody-stemmed, tendril climber. H 30ft (10m). Has 3-lobed leaves. Purple flowers, 3in (8cm) across, are borne from summer to fall. Z10–11 (min. 41°F/5°C)

P. violacea of gardens. See *P. ‘Amethyst’*.

P. vitifolia, syn. *P. sanguinea*. Evergreen, woody-stemmed, tendril climber; slender stems have fine, brown hairs. H to 15ft (5m). Has 3-lobed, lustrous leaves. In summer–fall, bears bright scarlet flowers, 5in (13cm) across, each with a short crown, banded red, yellow and white. Z12 (min. 61°F/16°C)

PATERSONIA

IRIDACEAE

Genus of evergreen, clump-forming, spring- and early summer-flowering, rhizomatous perennials. Frost tender. Requires full sun and light, well-drained soil. Leave undisturbed once planted. Propagate by seed in fall.

P. umbrosa. Evergreen, clump-forming, rhizomatous perennial. H to 3ft (1m), S 1ft (30cm). Has erect, basal leaves and tough flower stems bearing a succession of irislike, purple-blue flowers, 1¼–1½in (3–4cm) across, in summer. Z10–12 (min. 41°F/5°C)

PATRINIA

CAPRIFOLIACEAE

Genus of clump-forming perennials, grown for their flowers. Suitable for rock gardens and peat beds. Fully hardy. Needs a site in partial shade with moist soil. Propagate by division in spring or by seed in fall. Self-seeds freely.

P. triloba illus. p.275.

PAULOWNIA

PAULOWNIACEAE/SCROPHULARIACEAE

Genus of deciduous trees, grown for their large leaves and foxglovelike flowers, borne before the foliage emerges. Fully hardy, but flower buds and young growth of small plants may be damaged by very hard frosts. Requires full sun and fertile, moist but well-drained soil. Propagate by seed in fall or spring, or by root cuttings in winter.

P. fortunei. Deciduous, spreading tree. H and S 25ft (8m). Has large, ovate, mid-green leaves. In late spring, bears large, fragrant flowers, purple-spotted and white inside, pale purple outside. Z5–9.

FAST BLUE (‘Minfast’) starts flowering at only 3–4 years old.